

ICS 4111: Embedded Systems & IoT

Practical Exercise 3: Embedded System Design

Instructions:

In your assigned groups for the semester, you have been provided with the following components:

- A microcontroller
- A LED
- A breadboard
- One or two resistors
- A multimeter
- A potentiometer

Objective: Practical demonstration of Ohm's Law. For this exercise, you are expected to document your findings by responding to the reflection questions, taking images of your setups for each experiment. These can be presented as PDF document (s).

Experiment 1:

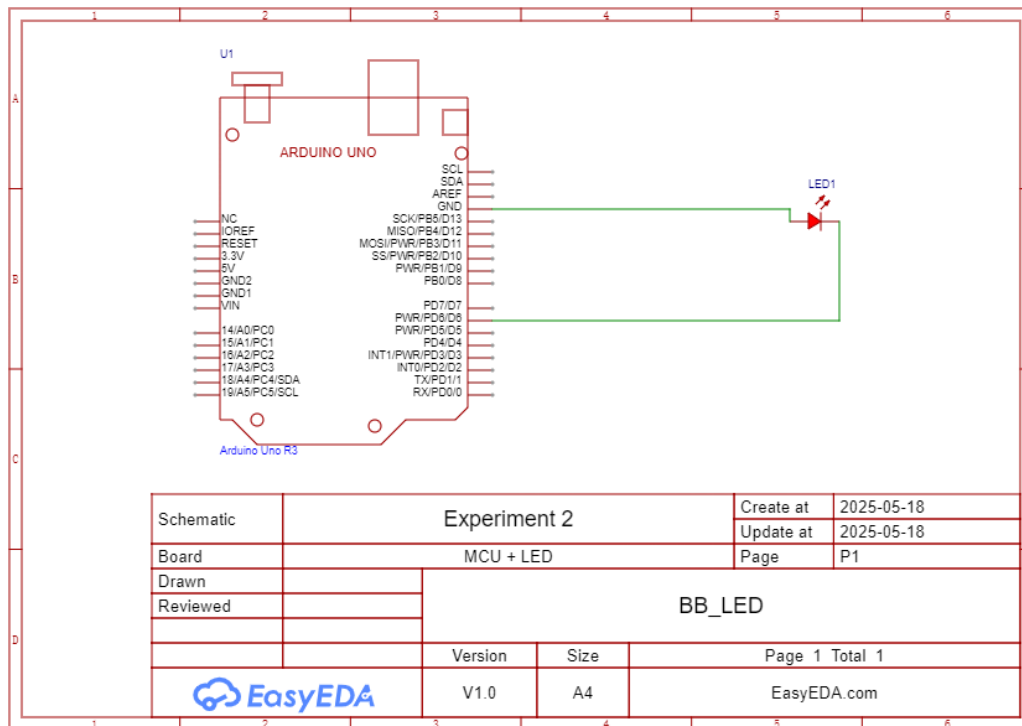
1. Pick one resistor, take a picture of it and determine its resistance value based on the colour code. Write down the resistance value.
2. Using a multimeter, measure the resistance value emitted by the resistor. Take a picture of this setup and write down the measured resistance.

You can check [this video](#) out or any other online resources on how to use a multimeter to measure voltage and resistance.

Experiment Reflection: Where does the difference in the resistance values arise from?

Experiment 2:

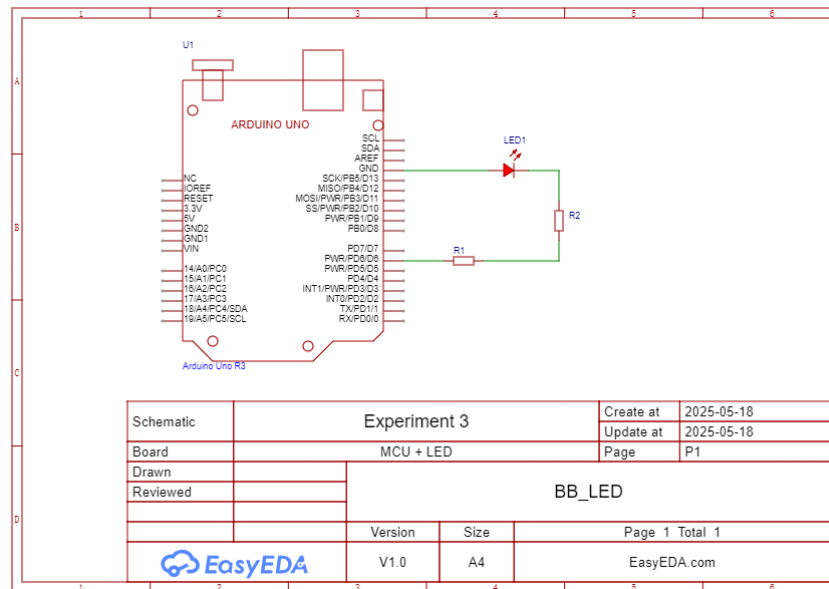
1. Using the breadboard, microcontroller, and LED, recreate the following schematic. Take a picture of the physical implementation.



2. In Arduino IDE, deploy the [Blink](#) program which is usually inbuilt in Arduino and upload to your board. *Congratulations you just created your first sketch! :)*
3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered.
4. Reconfigure the circuit you created in question 1 by adding a potentiometer to simulate [Pulse Width Modulation](#) (PWM) on the LED. Take 3 images of your circuit with the LED at 25% brightness, 50% brightness and 100% brightness. You can refer to these or any other suitable resources to guide on the connection: [Using PWM to Control the Light Intensity of a LED - Instructables](#); [Arduino LED Dimmer \(Potentiometer + PWM\)](#)

Experiment 3:

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in series. Take a picture of the physical implementation.

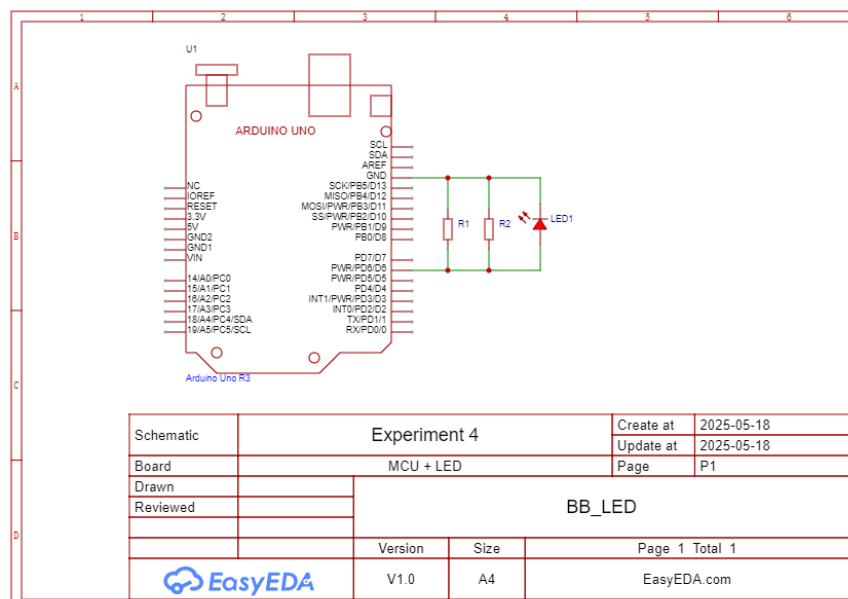


2. In Arduino IDE, re-deploy the [Blink](#) program and upload to your board.
3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

Experiment 4:

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in parallel. Take a picture of the physical implementation.



2. In Arduino IDE, re-deploy the [Blink](#) program and upload to your board.
3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

Prototyping Notice: In your free hours and during our physical classes this week, visit the SCES Makerspace Lab in the Engineering Labs Building to conduct your experiments. Schedule a prototyping session through [this link](#) and forward the invite to the instructor (avikiru@strathmore.edu) and lab technologists: (jintonjira@strathmore.edu, mngunju@strathmore.edu & amehta@strathmore.edu) to facilitate access to the lab. Please note, **no lab access will be granted without an invite email**. Ensure to read the lab rules as provided on e-learning.

Submission Instructions

Once done, publish your work to your GitHub account by following these steps:

1. One member of the group to access their GitHub account and navigate to the **Circuit-Design** repository.
2. Create a new branch called **Embedded-Design** and upload all documents used for this exercise. {Ensure that as you create the branch, there is no content from the Practical Exercise 2}
3. Since this is a group assignment, include evidence of groupwork in the repository. This evidence can be a cool team photo or minutes taken while working on the assignment. For photos, **faces of all members who worked on the assignment should be visible** whether the discussion was held physically or online. **Absent team members will not receive a teamwork score (50% of final score).**
4. After confirming everything is on the GitHub repo, copy the link to the repo from your address bar as *github.com/AllanVikiru/Circuit-Design/tree/Embedded-Design* and submit this link on e-learning before the due date. (confirm the appropriate due dates for your group on e-learning).

Feel free to use any available resources for your work, however the rules against plagiarism apply. **Do not directly use ChatGPT or any other AI content generators to develop your work**, this will be penalised strictly when detected.